



ABDELHAK BOURDIM

Senior Embedded Software Engineer
C/C++ • RTOS • Embedded Linux

CONTACT

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🔗 workshop-diy.org

LANGUAGES

French ●●●●● *Fluent*

English ●●●●● *Professional*

Arabic ●●●●● *Fluent*

STANDARDS & METHODS

- DO-178
- MISRA C
- IEC 62304
- ISO 15118 (V2G)
- ISO 14443 (Smart Card)
- ISO 15693 (RFID)
- OCPP
- V-Model
- Agile / Scrum

SKILLS

LANGUAGES
C, C++, Python, Bash, Assembly, Perl, Awk

OS / RTOS
Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel

MICROCONTROLLERS
ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto, Arduino

PROTOCOLS / BUS
CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP

STACKS
TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer

SECURITY / PAYMENT
Smart Card, RFID Mifare, DES, RC4, USB CCID

TOOLS
Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx, TouchGFX

MANAGEMENT

PROFILE

Embedded software specialist — firmware, middleware and application layers, bare metal, RTOS or Embedded Linux — for critical and industrial systems, with **over 20 years of experience** in driver development, communication stacks (CAN, BLE, USB, TCP/IP), RTOS and Embedded Linux on ARM Cortex, STM32, Microchip and NXP architectures.

Delivered certified embedded solutions for **aerospace** (Safran, Thales — DO-178), **medical** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT cybersecurity** (Enedis — Linky smart meters), **payment systems** (Cartadis) and **V2G smart charging** (E2-CAD — ISO 15118). MISRA C compliant development. Founder of **Workshop-DIY**, an educational robotics non-profit having led 18+ workshops for youth (Arduino, ESP32, AI, 3D printing).

PROFESSIONAL EXPERIENCE

E2-CAD — Eragny

May 2025 – Mar. 2026

EV Charging Station — Iotecha IP

Product takeover and test bench setup

- Reverse engineering and source code analysis (static and dynamic architecture)
- Embedded code review on 3 boards (Microchip PIC, STM32), setup of debug and programming tools (ST-Link, nsprog)
- Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, frame extraction with Saleae)
- Python extension development and real-time HTML web report generation
- Test bench design and setup: Chargebyte tools integration, EV simulator
- Charging station debug and commissioning (PLC, HomePlug Green PHY, ISO 15118, OCPP)

Env: Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, Chargebyte, RS232, I2C, CAN, TCP/IP, Saleae, HackRF One, SSH, SCP, Wireshark, Python, Bash, HTML

Safran — Créteil

Mar. 2024 – May 2025

SSPC Project — Solid State Power Controller

Low-level driver updates

- Development and update of low-level C drivers for the SSPC system
- Implementation of logging tools for diagnostics and exchange traceability
- Writing and execution of on-target validation tests with Lauterbach TRACE32
- Bash/Awk scripting for test process automation

Env: C, Lauterbach TRACE32, Bash, Awk

Asterios Technologies — Massy

Aug. 2023 – Feb. 2024

T1042SOM Project — Software Tests

Software test procedure writing

- C development of validation tests for the SOM module (T1042 processor)
- Processor errata analysis and software workaround identification
- Test documentation writing compliant with project requirements

Env: C, Bash, Python, Lauterbach TRACE32, Polarion

Thales — Gennevilliers

May 2022 – Jul. 2023

LPM + N-MMR Projects — Defense Radio

Mission preparation software & New Multi Mode Receiver

- Application layer development under Embedded Linux in C/C++ for 2 projects
- Configuration management and automation scripts (Bash, Curl, Tshark)
- Development and execution of HLT (High Level Tests)
- Bug fixing and network protocol analysis (Websockets)

Env: Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

Valeo — Créteil / Egypt

Mar. – Apr. 2022

Test Audit — Valeo Egypt

Unit test, integration test and tools audit

- Audit of test processes and tools (NI Mastre, Vector CANoe)
- Automation and execution time improvement proposals
- Client/system requirements review, audit report writing

Env: NI Mastre, Vector CANoe

Arkamys — Levallois-Perret

Jul. 2021 – Feb. 2022

GStreamer Framework — Embedded Audio

Audio plugins, unit & integration tests, End-to-End tests

- GStreamer plugin and CAN driver (ELM327) development, diagnostic tools
- Unit & integration testing (Valgrind), build system Buildroot

Env: Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

INDUSTRY SECTORS

- ✈ Aerospace — Safran, Thales, Zodiac
- 🏥 Medical — GE Healthcare, Sorin
- 🚗 Automotive — Valeo, Arkamys
- ⚡ Energy — Enedis (Linky)
- ⚡ EV Charging / V2G — E2-CAD
- 📶 IoT / M2M — Vertical M2M
- 💳 Payment — Cartadis, Photomaton, BCM

KEY STRENGTHS

- ✓ Embedded Linux & RTOS
- ✓ Embedded cybersecurity (pentest, audit)
- ✓ Embedded software development
- ✓ Communication protocols
- ✓ Testing & validation
- ✓ Diagnostics & debugging

NON-PROFIT PROJECT

Workshop-DIY

FOUNDER — CHELLES (77)

Robotics & STEM workshops for youth (ages 6-12+)
ESP32 robot kit design (WiFi, BLE, servos, sensors)
18+ workshops: Arduino, micro:bit, AI, 3D printing, laser
workshop-diy.org

EDUCATION & TECH WATCH

- Electronic & Computer Systems Design Engineer
AFPA, Champs-sur-Marne
- Biomedical Instrumentation (Postgraduate)
Tenon Hospital, Paris
- Electronics Engineer, Communication
Boumerdès, Algeria

TECHNOLOGY WATCH

- ▶ Embedded cybersecurity & IoT pentest
- ▶ V2G / ISO 15118 / Smart Charging
- ▶ SDR & RF protocol analysis
- ▶ Embedded AI & Edge Computing

PROFESSIONAL EXPERIENCE (CONTINUED)

Thales — Vélizy-Villacoublay

Nov. 2020 – Jun. 2021

ATO — Autonomous Train Operation

IPC communication & socket programming

- ▶ Application migration from 32-bit to 64-bit on Embedded Linux platform
- ▶ IPC communication protocol definition and implementation
- ▶ Network diagnostic tools development and integration testing
- ▶ Frame analysis with Wireshark, Tshark, Pscapy and Pyshark

Env: Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Pscapy, Python, Jira

PK Vitality — Paris

Jun. – Sep. 2020

K'Watch CGM — Continuous Glucose Monitor

BLE drivers, FreeRTOS, STM32 TouchGFX

- ▶ HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- ▶ STM32 TouchGFX touchscreen, FreeRTOS, low power mode
- ▶ In vitro testing and Doxygen documentation (IEC 62304 compliant)

Env: STM32L4S9, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen

Zodiac Aerospace (Safran) — Montreuil / Vancouver, Canada
GENOME / CORAC / SSPC A350 AC9

Feb. 2017 – Mar. 2020

Low-level drivers, diagnostics, application layers

- ▶ Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- ▶ uAFDX to CAN gateway implementation and diagnostic functions
- ▶ SSPC A350 application layer development and optimization
- ▶ Validation testing and aerospace standards-compliant documentation
- ▶ Flight test campaign at UBC Vancouver, Canada (DO-178 certification)

Env: Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C

Sorin — Clamart

Nov. 2014 – Apr. 2015

Platinum — BLE Module for Pacemaker

Bluetooth integration feasibility study for cardiac implant

- ▶ Bluetooth Smart module validation, Serial Port Service Profile development
- ▶ Smartphone communication protocol development and Android application
- ▶ Hardware interface design and sample preparation for implantation

Env: Cortex M0, Keil, C, Android (Eclipse)

General Electric Healthcare — Buc / USA

2012 – 2014

ZYX — X-Ray Generator

CANOpen integration on RCB and CORTYX boards

- ▶ CANOpen stack integration (slave/master), CAN driver and service development
- ▶ CMX-RTX kernel porting, file management, TCL test scripts
- ▶ USA team technical support and on-site visits in the United States

Env: TI TMS320, XE167, ARM Cortex A8, VxWorks, C, C++

PREVIOUS EXPERIENCE

Enedis — Nanterre

May 2015 – Sep. 2016

Linky Smart Meters — Smart Grids

Cybersecurity and Linky concentrator audit

- ▶ Linky concentrator security diagnostics and audit
- ▶ Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- ▶ Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)
- ▶ Log analysis and batch firmware update tool development
- ▶ Code audit and review with manufacturers (Sonar, PC Lint)

Env: Kali Linux, Embedded Linux, Nmap, Wireshark, Tshark, Scapy, SDR, Saleae, Python, Bash, Arduino, Raspberry Pi

Photomaton / BCM / Vertical M2M / PGS — Paris

2010 – 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

Cartadis — Fontenay-sous-Bois

Oct. 2001 – Dec. 2009

8 years — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)